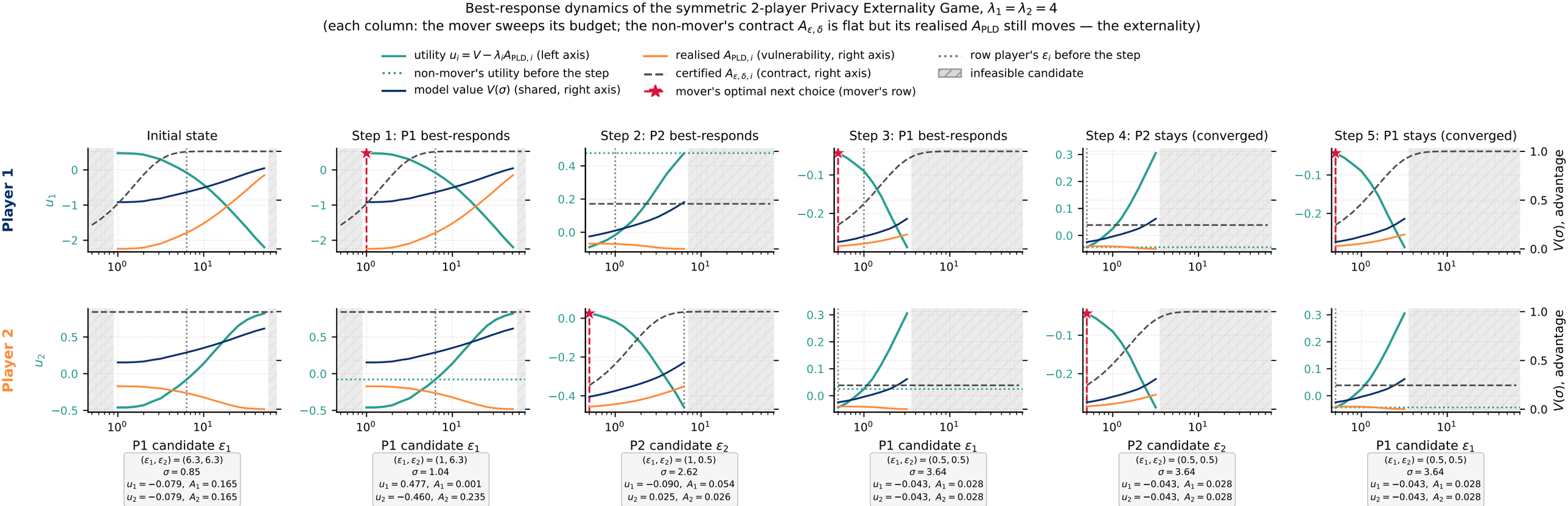
Best-response dynamics of the symmetric 2-player Privacy Externality Game, $\lambda_1 = \lambda_2 = 4$
(each column: the mover sweeps its budget; the non-mover's contract $A_{\varepsilon,\delta}$ is flat but its realised A_{PLD} still moves — the externality)
utility $u_i = V - \lambda_i A_{\mathrm{PLD},i}$ (left axis)
non-mover's utility before the step
model value $V(\sigma)$ (shared, right axis)
realised $A_{\mathrm{PLD},i}$ (vulnerability, right axis)
certified $A_{\varepsilon,\delta,i}$ (contract, right axis)
mover's optimal next choice (mover's row)
row player's ε_i before the step
infeasible candidate
Initial state
Step 1: P1 best-responds
Step 2: P2 best-responds
Step 3: P1 best-responds
Step 4: P2 stays (converged)
Step 5: P1 stays (converged)
Player 1
Player 2
u_1
u_2
$V(\sigma)$, advantage
P1 candidate ε_1
P1 candidate ε_1
P2 candidate ε_2
P1 candidate ε_1
P2 candidate ε_2
P1 candidate ε_1
$(\varepsilon_1, \varepsilon_2) = (6.3, 6.3)$
$\sigma = 0.85$
$u_1 = -0.079,\ A_1 = 0.165$
$u_2 = -0.079,\ A_2 = 0.165$
$(\varepsilon_1, \varepsilon_2) = (1, 6.3)$
$\sigma = 1.04$
$u_1 = 0.477,\ A_1 = 0.001$
$u_2 = -0.460,\ A_2 = 0.235$
$(\varepsilon_1, \varepsilon_2) = (1, 0.5)$
$\sigma = 2.62$
$u_1 = -0.090,\ A_1 = 0.054$
$u_2 = 0.025,\ A_2 = 0.026$
$(\varepsilon_1, \varepsilon_2) = (0.5, 0.5)$
$\sigma = 3.64$
$u_1 = -0.043,\ A_1 = 0.028$
$u_2 = -0.043,\ A_2 = 0.028$
$(\varepsilon_1, \varepsilon_2) = (0.5, 0.5)$
$\sigma = 3.64$
$u_1 = -0.043,\ A_1 = 0.028$
$u_2 = -0.043,\ A_2 = 0.028$
$(\varepsilon_1, \varepsilon_2) = (0.5, 0.5)$
$\sigma = 3.64$
$u_1 = -0.043,\ A_1 = 0.028$
$u_2 = -0.043,\ A_2 = 0.028$